RAI-818: INTELLIGENT MOBILE ROBOTICS

FastSLAM
- Introduction
- Key Steps
- Pseudo Code
- The Importance Weight

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:
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Particle Filter

- Non-parametric recursive Bayes filter
- Posterior is represented by a set of weighted samples
- Can model arbitrary distributions
- Works well in low-dimensional spaces
- 3-Step procedure
  - Sampling from proposal
  - Importance Weighting
  - Resampling

Fast SLAM

- Introduction
  - Particle Filter

Particle Filter

1. Sample the particles from the proposal distribution

$$x_t^{[j]} \sim \pi(x_t \mid \ldots)$$

2. Compute the importance weights

$$w_t^{[j]} = \frac{target(x_t^{[j]})}{proposal(x_t^{[j]})}$$

3. Resampling: Draw sample $\boldsymbol{i}$ with probability $\boldsymbol{w_t^{[i]}}$ and repeat $\boldsymbol{J}$ times

Defining futures

Fast SLAM

- Introduction
  - Particle Filter

Particle Representation

1. A set of weighted samples

$$\mathcal{X} = \left\{ \left\langle x^{[i]}, w^{[i]} \right\rangle \right\}_{i=1,\ldots,N}$$

2. Think of a sample as one hypothesis about the state

3. For feature-based SLAM:

$$x = (\underbrace{x_{1:t}}_{poses}, \underbrace{m_{1,x}, m_{1,y}, \ldots, m_{M,x}, m_{M,y}}_{landmarks})^T$$

Defining futures

Fast SLAM

Dimensionality Problem

1. Particle filters are effective in low dimensional spaces. The likely regions of the state space need to be covered with samples.
2. Higher dimensions → more samples.

$$x = \underbrace{(x_{1:t}, m_{1,x}, m_{1,y}, \ldots, m_{M,x}, m_{M,y})^T}_{\textit{High Dimensions}}$$

3. Can We Exploit Dependencies Between the Different Dimensions of the State Space?

$$x_{1:t}, m_1, \ \ldots, m_M$$

4. If We Know the Poses of the Robot, Mapping is Easy!

$$\underline{x_{1:t}}, \underline{m_1, \ \ldots, m_M}$$

5. If we use the particle set only to model the robot's path, each sample is a path hypothesis. For each sample, we can compute an individual map of landmarks.

Defining futures

Rao-Blackwellization

1. Factorization to exploit dependencies between variables:

$$p(a, b) = p(b \mid a)\, p(a)$$

2. If $p(b| a)$ can be computed in closed form, represent only $p(b| a)$ with $p(a)$ samples and compute $p(b| a)$ for every sample

3. Factorization of the SLAM posterior

poses *map* *observations* *movements*

$$p(x_{0:t}, m_{1:M} \mid z_{1:t}, u_{1:t}) =$$

$$p(x_{0:t} \mid z_{1:t}, u_{1:t})\; p(m_{1:M} \mid x_{0:t}, z_{1:t})$$

path posterior *map posterior*

4. How to compute the map posterior efficiently?

Defining futures

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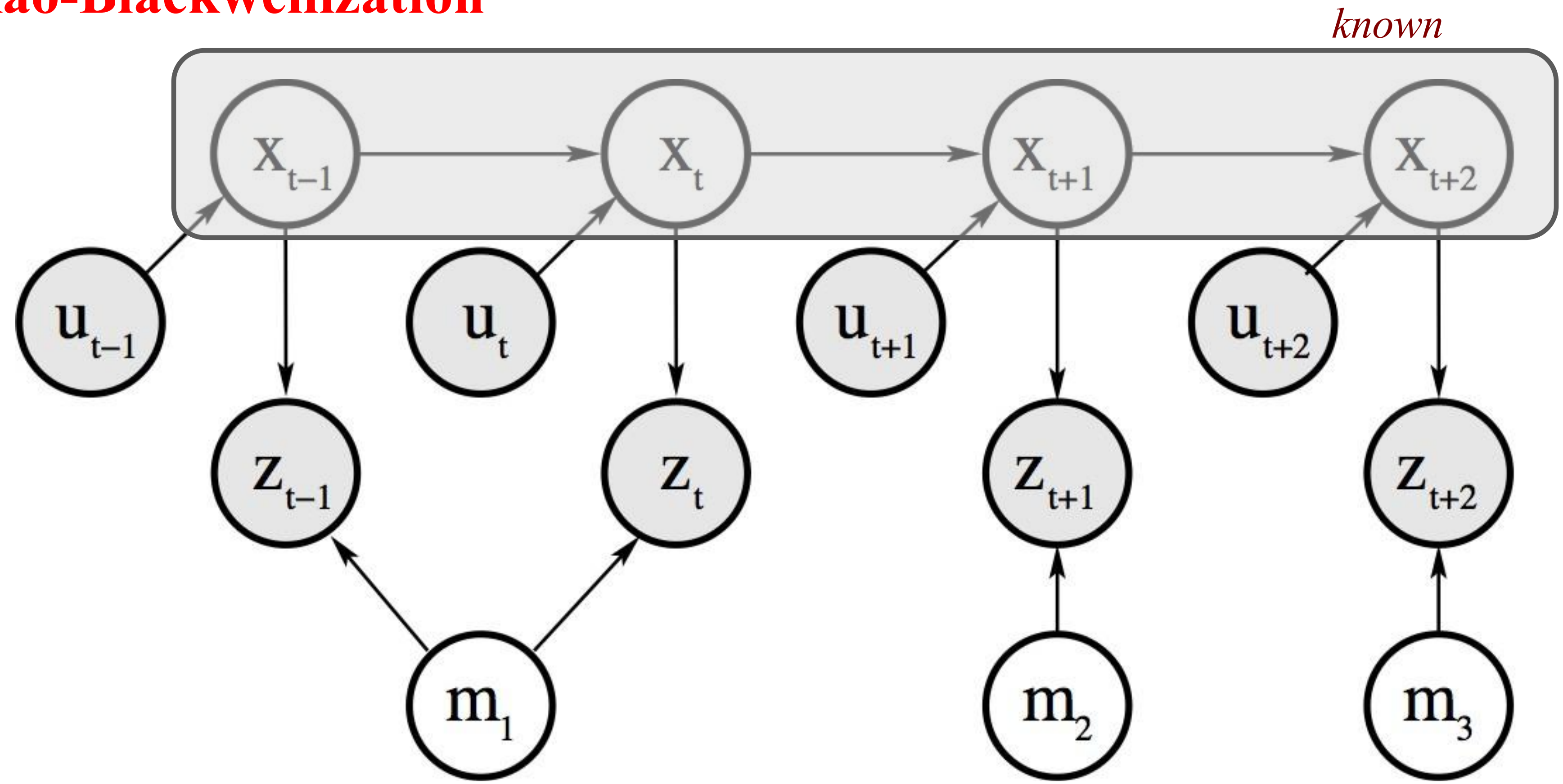
Fast SLAM

Rao-Blackwellization



Landmarks are Conditionally Independent Given the Poses

Landmark variables are all disconnected (i.e. independent) given the robot's path

Rao-Blackwellization

1. Factorization of the SLAM posterior

Landmarks are conditionally independent given the poses

$$p(x_{0:t}, m_{1:M} \mid z_{1:t}, u_{1:t}) =$$

$$p(x_{0:t} \mid z_{1:t}, u_{1:t})\; p(m_{1:M} \mid x_{0:t}, z_{1:t})$$

$$p(x_{0:t} \mid z_{1:t}, u_{1:t}) \prod_{i=1}^{M} p(m_i \mid x_{0:t}, z_{1:t})$$

particle filter similar to MCL

2D EKFs

Defining futures

Modelling the Robot's Path

- Sample-based representation

$$p(x_{0:t} \mid z_{1:t}, u_{1:t})$$

- Each sample is a path hypothesis

$$x_0 \qquad x_1 \qquad x_2 \qquad \ldots$$

starting location, typically (0,0,0)

pose hypothesis at time t=1

- Past poses of a sample are not revised
- No need to maintain past poses in the sample set

Defining futures

Fast SLAM

- Introduction
- Fast SLAM
  - Basic Algorithm

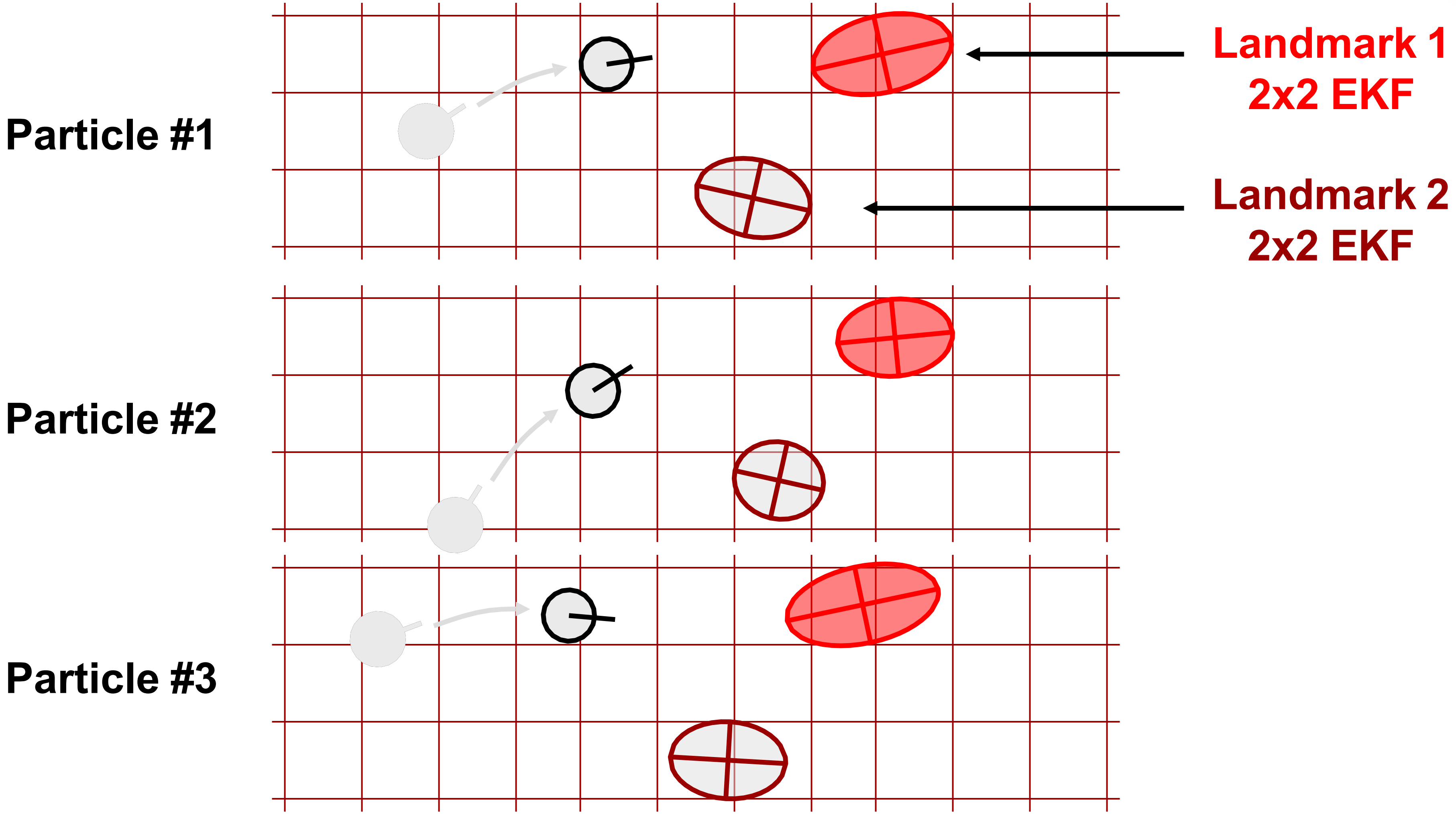
Modelling the Robot's Path

- Proposed by Montemerlo et al. in 2002
- Each landmark is represented by a 2x2 EKF
- Each particle therefore has to maintain M individual EKFs

Particle 1	x, y, θ	Landmark 1	Landmark 2	...	Landmark M
Particle 2	x, y, θ	Landmark 1	Landmark 2	...	Landmark M
⋮					
Particle N	x, y, θ	Landmark 1	Landmark 2	...	Landmark M

Defining futures

Fast SLAM – Action Update

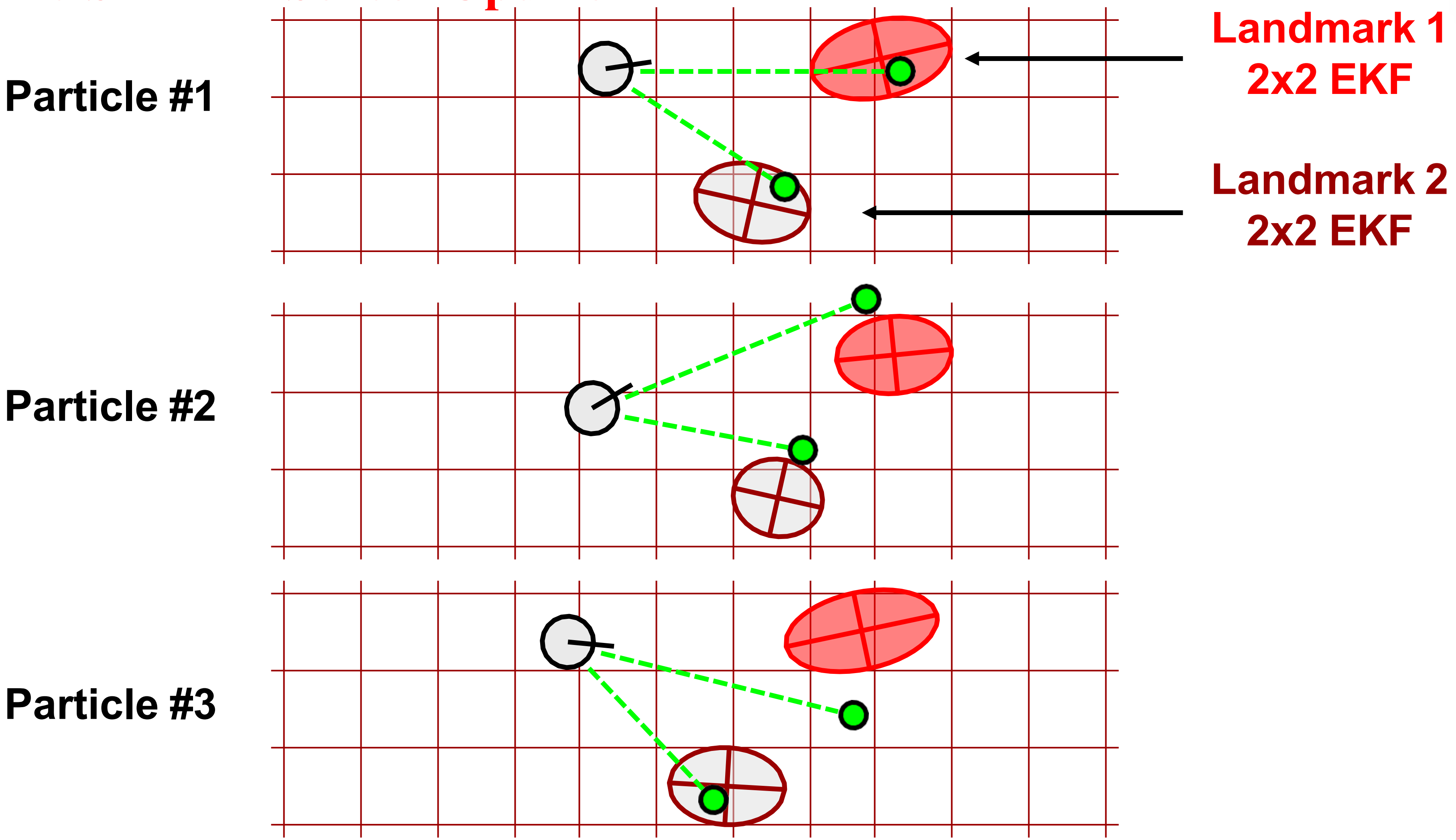


Defining futures

Fast SLAM

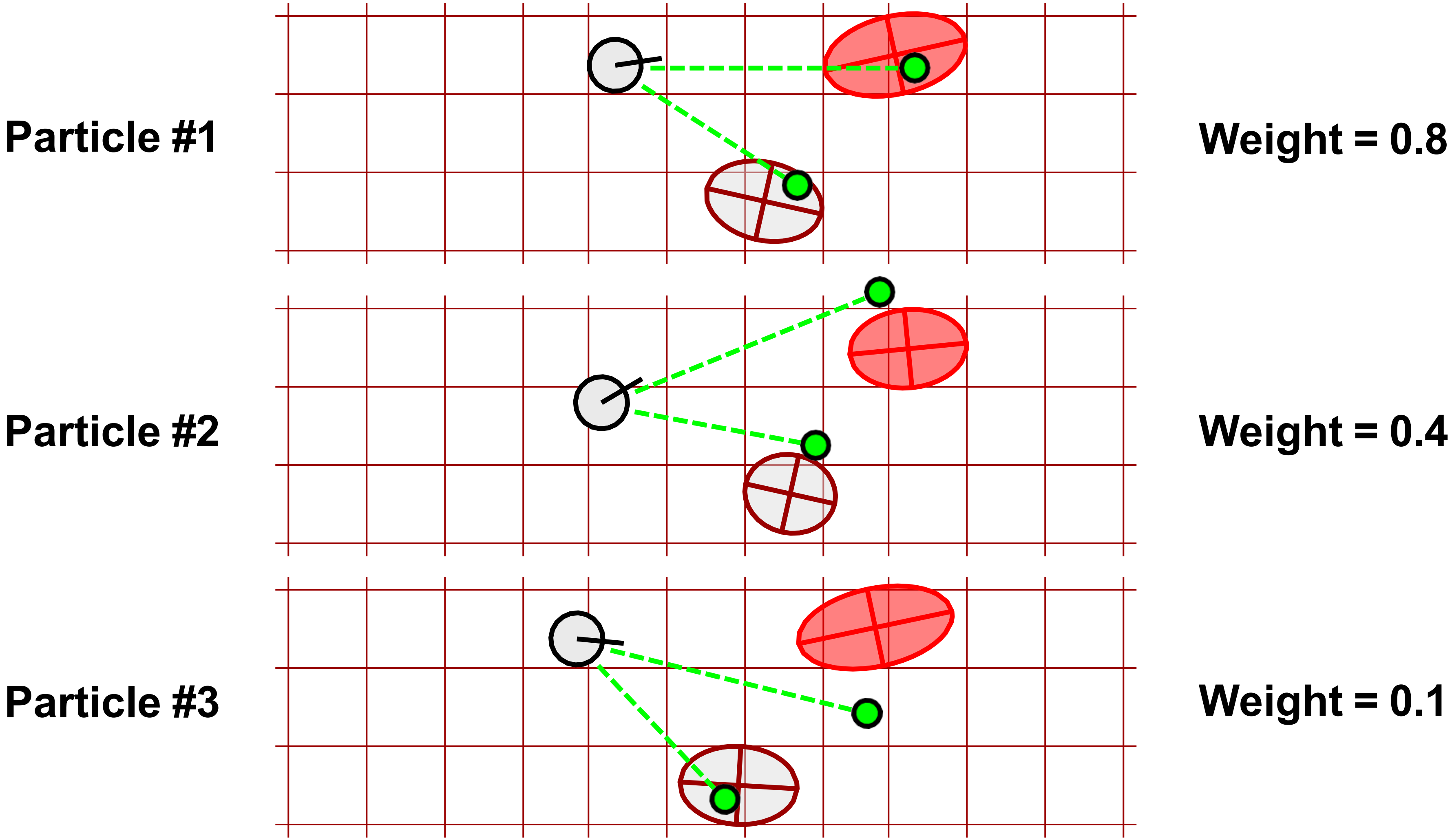
- Introduction
- Fast SLAM
  - Basic Algorithm
  - Action Update
  - Sensor Update

FastSLAM – Sensor Update



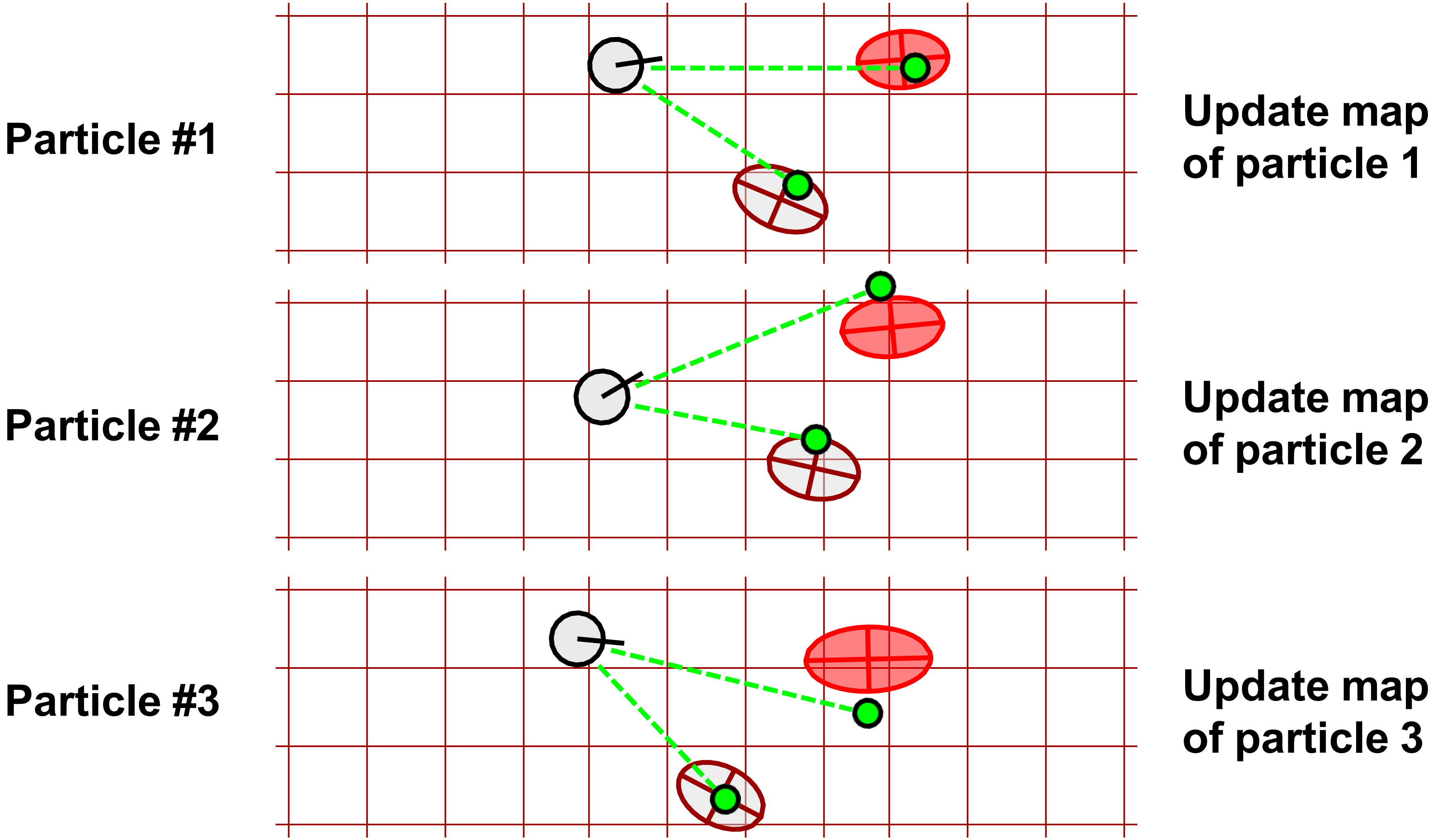
Defining futures

FastSLAM – Sensor Update



Defining futures

FastSLAM – Sensor Update



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Key Steps of Fast SLAM 1.0

- Extend the path posterior by sampling a new pose for each sample

$$x_t^{[k]} \sim p(x_t \mid x_{t-1}^{[k]}, u_t)$$

- Compute particle weight

$$w^{[k]} = |2\pi Q|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(z_t - \hat{z}^{[k]})^T \underbrace{Q^{-1}}_{\text{measurement covariance}} (z_t - \overbrace{\hat{z}^{[k]}}^{\text{exp. observation}})\right\}$$

- Update belief of observed landmarks (EKF update rule)
- Resample

FastSLAM 1.0 – Part 1

1: **FastSLAM1.0_known_correspondence**$(z_t, c_t, u_t, \mathcal{X}_{t-1})$:

2: *for* $k = 1$ *to* N *do* // *loop over all particles*

3: *Let* $\left\langle x_{t-1}^{[k]}, \left\langle \mu_{1,t-1}^{[k]}, \Sigma_{1,t-1}^{[k]} \right\rangle, \ldots \right\rangle$ *be particle* k *in* $\mathcal{X}_{t-1}$

4: $x_t^{[k]} \sim p(x_t \mid x_{t-1}^{[k]}, u_t)$ // *sample pose*

Defining futures

FastSLAM 1.0 – Part 1

FastSLAM1.0_known_correspondence$(z_t, c_t, u_t, \mathcal{X}_{t-1})$:

$\quad$ *for* $k = 1$ *to* N *do* $\qquad$ // *loop over all particles*

$\qquad$ *Let* $\left\langle x_{t-1}^{[k]}, \left\langle \mu_{1,t-1}^{[k]}, \Sigma_{1,t-1}^{[k]} \right\rangle, \ldots \right\rangle$ *be particle* k *in* $\mathcal{X}_{t-1}$

$\qquad$ $x_t^{[k]} \sim p(x_t \mid x_{t-1}^{[k]}, u_t)$ $\qquad$ // *sample pose*

$\qquad$ $j = c_t$ $\qquad$ // *observed feature*

$\qquad$ *if feature* j *never seen before*

$\qquad\quad$ $\mu_{j,t}^{[k]} = h^{-1}(z_t, x_t^{[k]})$ $\qquad$ // *initialize mean*

$\qquad\quad$ $H = h'(\mu_{j,t}^{[k]}, x_t^{[k]})$ $\qquad$ // *calculate Jacobian*

$\qquad\quad$ $\Sigma_{j,t}^{[k]} = H^{-1}\, Q_t\, (H^{-1})^T$ $\qquad$ // *initialize covariance*

$\qquad\quad$ $w^{[k]} = p_0$ $\qquad$ // *default importance weight*

$\qquad$ *else*

Defining futures

FastSLAM 1.0 – Part 2

- Introduction
- Fast LAM
- **FastSLAM 1.0**
  - Key Steps
  - **Pseudo Code**

$\quad\quad$ *else*

$\quad\quad\quad \langle \mu_{j,t}^{[k]}, \Sigma_{j,t}^{[k]} \rangle = \textit{EKF-Update}(\dots) \qquad //\ \textit{update landmark}$

$\quad\quad\quad w^{[k]} = |2\pi Q|^{-\frac{1}{2}} \ \exp\left\{ -\frac{1}{2}(z_t - \hat{z}^{[k]})^T Q^{-1} \ (z_t - \hat{z}^{[k]}) \right\}$

measurement cov. $\quad Q = H \ \Sigma_{j,t-1}^{[k]} \ H^T + Q_t \quad$ *exp. observation*

$\quad\quad$ *endif*

$\quad\quad$ *for all unobserved features* j' *do*

$\quad\quad\quad \langle \mu_{j',t}^{[k]}, \Sigma_{j',t}^{[k]} \rangle = \langle \mu_{j',t-1}^{[k]}, \Sigma_{j',t-1}^{[k]} \rangle \qquad //\ \textit{leave unchanged}$

$\quad\quad$ *endfor*

$\quad$ *endfor*

$\quad \mathcal{X}_t = \text{resample}\left(\left\langle x_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]} \right\rangle, \dots, w^{[k]} \right\rangle_{k=1,\dots,N} \right)$

$\quad$ *return* $\mathcal{X}_t$

Defining futures

FastSLAM 1.0 – Part 2 (long)

else

EKF update

$\hat{z}^{[k]} = h(\mu_{j,t-1}^{[k]}, x_t^{[k]})$ // *measurement prediction*

$H = h'(\mu_{j,t-1}^{[k]}, x_t^{[k]})$ // *calculate Jacobian*

$Q = H\ \Sigma_{j,t-1}^{[k]}\ H^T + Q_t$ // *measurement covariance*

$K = \Sigma_{j,t-1}^{[k]}\ H^T\ Q^{-1}$ // *calculate Kalman gain*

$\mu_{j,t}^{[k]} = \mu_{j,t-1}^{[k]} + K(z_t - \hat{z}^{[k]})$ // *update mean*

$\Sigma_{j,t}^{[k]} = (I - K\ H)\Sigma_{j,t-1}^{[k]}$ // *update covariance*

$w^{[k]} = |2\pi Q|^{-\frac{1}{2}}\ \exp\left\{-\frac{1}{2}(z_t - \hat{z}^{[k]})^T\ Q^{-1}\ (z_t - \hat{z}^{[k]})\right\}$ // *importance factor*

endif

for all unobserved features j' do

$\langle \mu_{j',t}^{[k]}, \Sigma_{j',t}^{[k]} \rangle = \langle \mu_{j',t-1}^{[k]}, \Sigma_{j',t-1}^{[k]} \rangle$ // *leave unchanged*

endfor

endfor

$\mathcal{X}_t = \text{resample}\left(\left\langle x_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]} \right\rangle, \ldots, w^{[k]} \right\rangle_{k=1,\ldots,N}\right)$

return $\mathcal{X}_t$

The Importance Weight

- The Weight is a Result From the Importance Sampling Principle
- Importance weight is given by the ratio of target and proposal in $x^{[k]}$
- See: importance sampling principle

$$w^{[k]} = \frac{\text{target}(x^{[k]})}{\text{proposal}(x^{[k]})}$$

Defining futures

The Importance Weight

- The target distribution is

$$p(x_{1:t} \mid z_{1:t}, u_{1:t})$$

- The proposal distribution is

$$p(x_{1:t} \mid z_{1:t-1}, u_{1:t})$$

- Proposal is used step-by-step

$$p(x_{1:t} \mid z_{1:t-1}, u_{1:t}) = \underbrace{p(x_t \mid x_{t-1}, u_t)}_{\text{from } \mathcal{X}_{t-1} \text{ to } \bar{\mathcal{X}}_t} \underbrace{p(x_{1:t-1} \mid z_{1:t-1}, u_{1:t-1})}_{\mathcal{X}_{t-1}}$$

Defining futures

The Importance Weight

$$
\begin{aligned}
w^{[k]} &= \frac{\text{target}(x^{[k]})}{\text{proposal}(x^{[k]})} \\
&= \frac{p(x_{1:t}^{[k]} \mid z_{1:t}, u_{1:t})}{p(x_t^{[k]} \mid x_{t-1}, u_t)\; p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})} \\
&= \frac{\eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1})\; p(x_t \mid x_{t-1}^{[k]}, u_t)}{p(x_t^{[k]} \mid x_{t-1}^{[k]}, u_t)} \\
&\quad \frac{p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})}{p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})} \\
&= \eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1})
\end{aligned}
$$

Defining futures

The Importance Weight

$$
\begin{aligned}
w^{[k]} &= \frac{\text{target}(x^{[k]})}{\text{proposal}(x^{[k]})} \\
&= \frac{p(x_{1:t}^{[k]} \mid z_{1:t}, u_{1:t})}{p(x_t^{[k]} \mid x_{t-1}, u_t)\; p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})} \\
&= \frac{\eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1})\; p(x_t \mid x_{t-1}^{[k]}, u_t)}{p(x_t^{[k]} \mid x_{t-1}^{[k]}, u_t)} \\
&\quad \frac{p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})}{p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})} \\
&= \eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1})
\end{aligned}
$$

Bayes rule + factorization

Defining futures

The Importance Weight

$$
\begin{aligned}
w^{[k]} &= \frac{\text{target}(x^{[k]})}{\text{proposal}(x^{[k]})} \\
&= \frac{p(x_{1:t}^{[k]} \mid z_{1:t}, u_{1:t})}{p(x_t^{[k]} \mid x_{t-1}, u_t)\; p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})} \\
&= \frac{\eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1})\; \cancel{p(x_t \mid x_{t-1}^{[k]}, u_t)}}{\cancel{p(x_t^{[k]} \mid x_{t-1}^{[k]}, u_t)}} \\
&\quad \frac{\cancel{p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})}}{\cancel{p(x_{1:t-1}^{[k]} \mid z_{1:t-1}, u_{1:t-1})}} \\
&= \eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1})
\end{aligned}
$$

Defining futures

The Importance Weight

- Integrating over the pose of the observed landmark leads to

$$
\begin{aligned}
w^{[k]} & \\
&= \eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1}) \\
&= \eta \int p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1}, m_j)\, p(m_j \mid x_{1:t}^{[k]}, z_{1:t-1})\, dm_j
\end{aligned}
$$

Defining futures

The Importance Weight

- Integrating over the pose of the observed landmark leads to

$$
\begin{aligned}
w^{[k]} & \\
&= \eta \, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1}) \\
&= \eta \int p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1}, m_j) \, p(m_j \mid x_{1:t}^{[k]}, z_{1:t-1}) \, dm_j \\
&= \eta \int p(z_t \mid x_t^{[k]}, m_j) \; p(m_j \mid x_{1:t-1}^{[k]}, z_{1:t-1}) \; dm_j
\end{aligned}
$$

Defining futures

The Importance Weight

- Integrating over the pose of the observed landmark leads to

$$
\begin{aligned}
w^{[k]} & \\
&= \eta\, p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1}) \\
&= \eta \int p(z_t \mid x_{1:t}^{[k]}, z_{1:t-1}, m_j)\, p(m_j \mid x_{1:t}^{[k]}, z_{1:t-1})\, dm_j \\
&= \eta \int \underbrace{p(z_t \mid x_t^{[k]}, m_j)}_{\mathcal{N}(z_t;\hat{z}^{[k]},Q_t)}\ \underbrace{p(m_j \mid x_{1:t-1}^{[k]}, z_{1:t-1})}_{\mathcal{N}(m_j;\mu_{j,t-1}^{[k]},\Sigma_{j,t-1}^{[k]})}\, dm_j
\end{aligned}
$$

Defining futures

The Importance Weight

- This leads to

$$
\begin{aligned}
w^{[k]} & \\
&= \eta \, p(z_t \mid x^{[k]}_{1:t}, z_{1:t-1}) \\
&= \eta \int p(z_t \mid x^{[k]}_{1:t}, z_{1:t-1}, m_j) \, p(m_j \mid x^{[k]}_{1:t}, z_{1:t-1}) \, dm_j \\
&= \eta \int \underbrace{p(z_t \mid x^{[k]}_t, m_j)}_{\mathcal{N}(z_t;\hat{z}^{[k]},Q_t)} \; \underbrace{p(m_j \mid x^{[k]}_{1:t-1}, z_{1:t-1})}_{\mathcal{N}(m_j;\mu^{[k]}_{j,t-1},\Sigma^{[k]}_{j,t-1})} \, dm_j
\end{aligned}
$$

$$Q = H \, \Sigma^{[k]}_{j,t-1} \, H^T + Q_t$$

$$w^{[k]} \simeq |2\pi Q|^{-\frac{1}{2}} \exp\left\{ -\frac{1}{2} (z_t - \hat{z}^{[k]})^T Q^{-1} (z_t - \hat{z}^{[k]}) \right\}$$

measurement covariance (pose uncertainty of the landmark estimate plus measurement noise)

Fast SLAM

- Introduction
- Fast LAM
- **FastSLAM 1.0**
  - Key Steps
  - Pseudo Code
  - **The Importance Weight**


```
else
    ⟨μ_{j,t}^{[k]}, Σ_{j,t}^{[k]}⟩ = EKF-Update(...)    // update landmark
```

$$\boxed{w^{[k]} = |2\pi Q|^{-\frac{1}{2}} \ \exp\left\{-\frac{1}{2}(z_t - \hat{z}^{[k]})^T Q^{-1} \ (z_t - \hat{z}^{[k]})\right\}}$$

```
endif
for all unobserved features j' do
    ⟨μ_{j',t}^{[k]}, Σ_{j',t}^{[k]}⟩ = ⟨μ_{j',t-1}^{[k]}, Σ_{j',t-1}^{[k]}⟩    // leave unchanged
endfor
endfor
```

$$\mathcal{X}_t = \text{resample}\left(\left\langle x_t^{[k]}, \left\langle \mu_{1,t}^{[k]}, \Sigma_{1,t}^{[k]} \right\rangle, \ldots, w^{[k]} \right\rangle_{k=1,\ldots,N}\right)$$

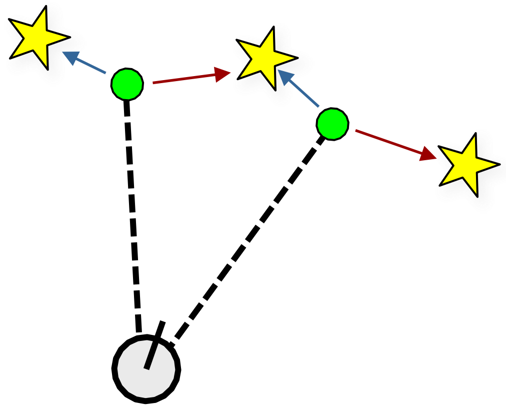
return $\mathcal{X}_t$

Defining futures

Fast SLAM

Data Association Problem

- Which observation belongs to which landmark?

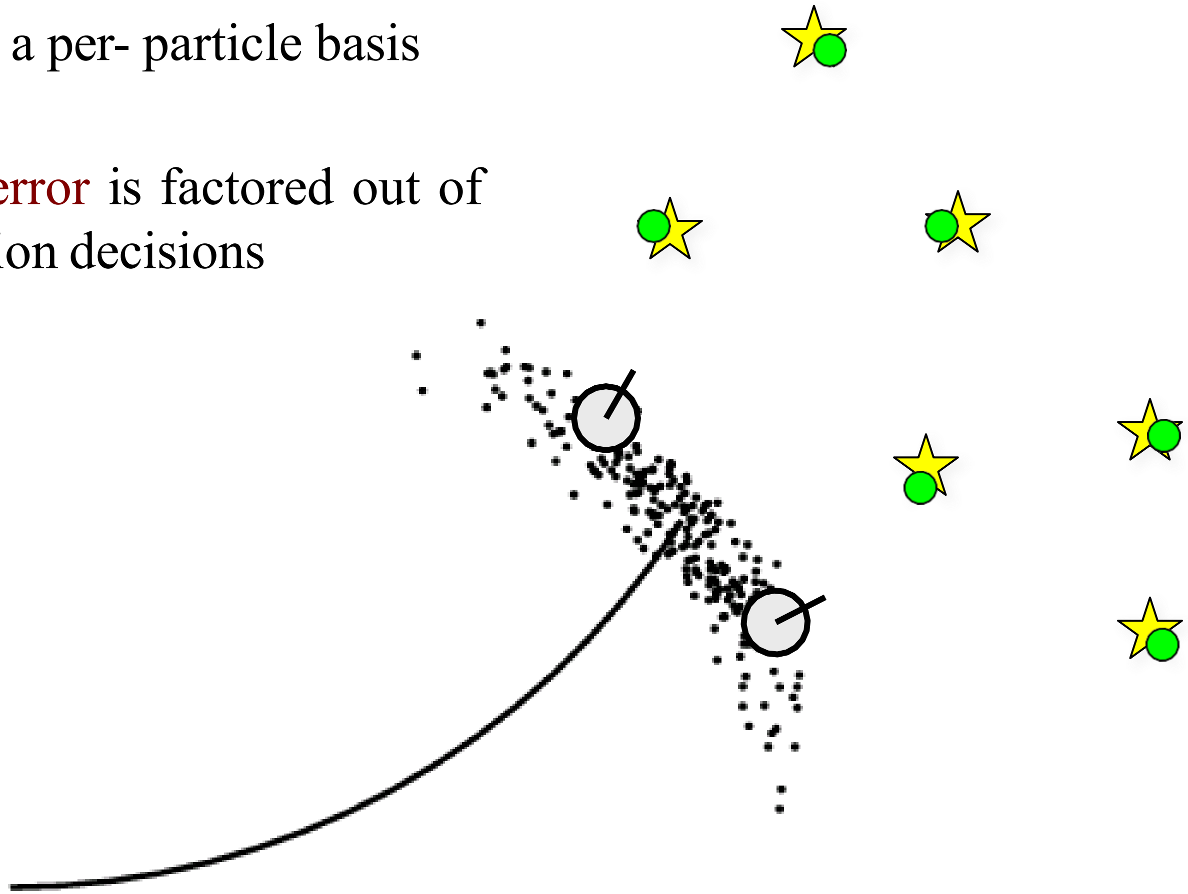


- More than one possible association
- Potential data associations depend on the pose of the robot

Fast SLAM

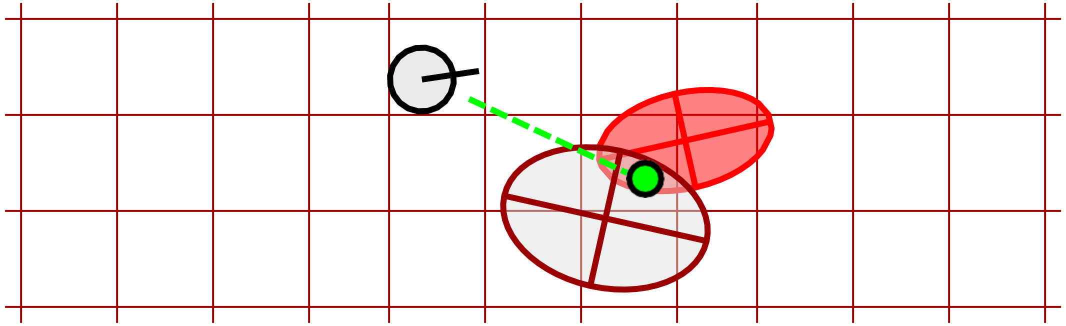
Data Association Problem

- Particles Support for Multi- Hypotheses Data Association
- Decisions on a per- particle basis
- Robot pose error is factored out of data association decisions



Fast SLAM

Per-Particle Data Association



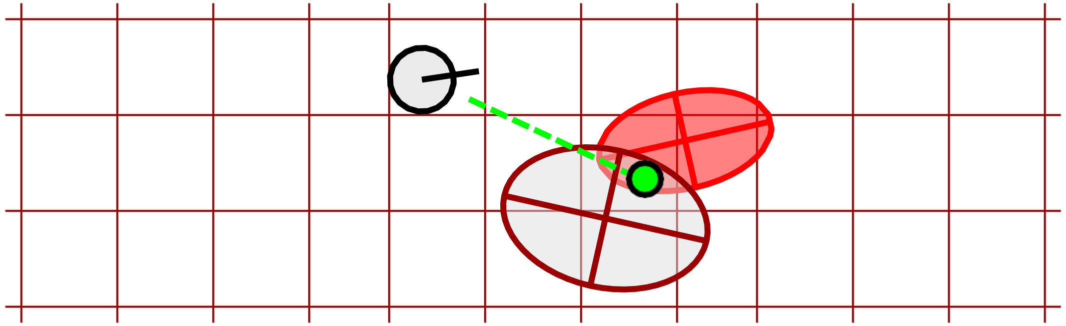
Was the observation generated by the red or by the brown landmark?

P(observation|red) = 0.3

P(observation|brown) = 0.7

Fast SLAM

Per-Particle Data Association



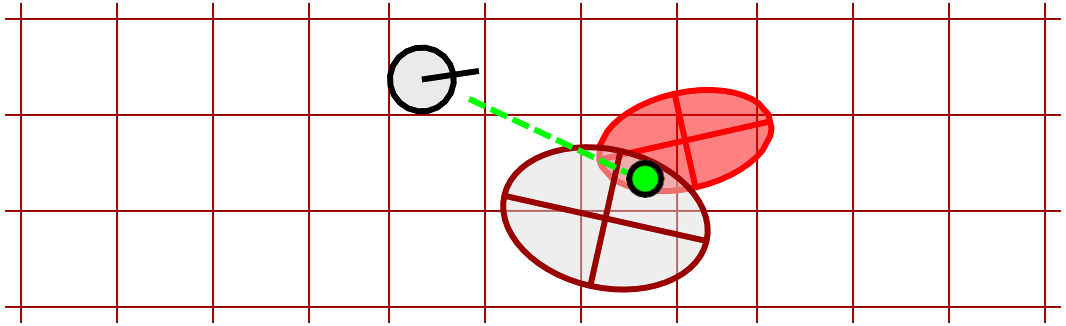
Was the observation generated by the red or by the brown landmark?

P(observation | red) = 0.3 P(observation | brown) = 0.7

- Two options for per-particle data association
  - Pick the most probable match
  - Pick an random association weighted by the observation likelihoods
- If the probability for an assignment is too low, generate a new landmark

Fast SLAM

Per-Particle Data Association



Was the observation generated by the red or by the brown landmark?

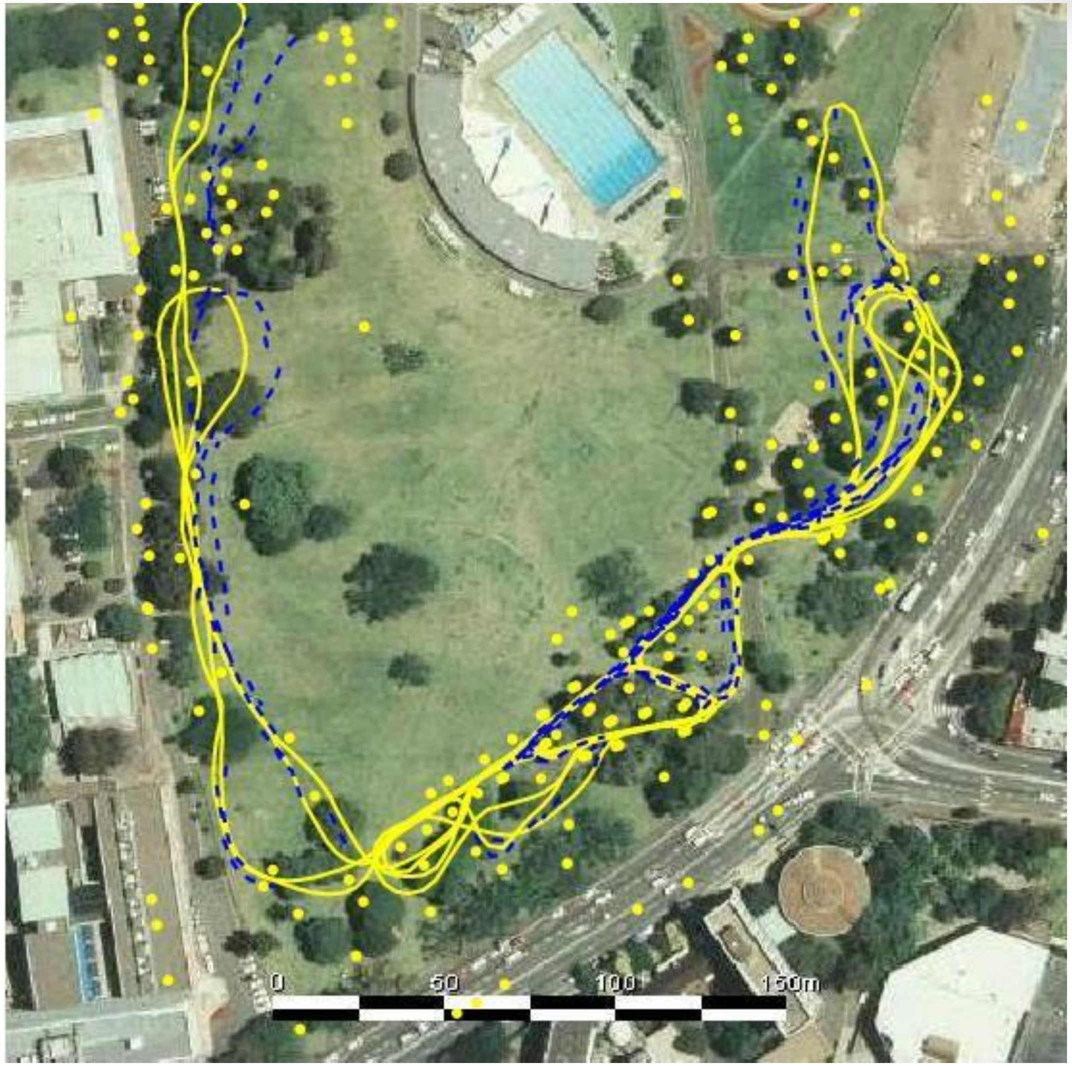
- Multi-modal belief
- Pose error is factored out of data association decisions
- Simple but effective data association
- Big advantage of FastSLAM over EKF

Defining futures

Fast SLAM

Results – Victoria Park

- 4 km traverse
- < 2.5 m RMS position error
- 100 particles



Blue = GPS
Yellow = FastSLAM

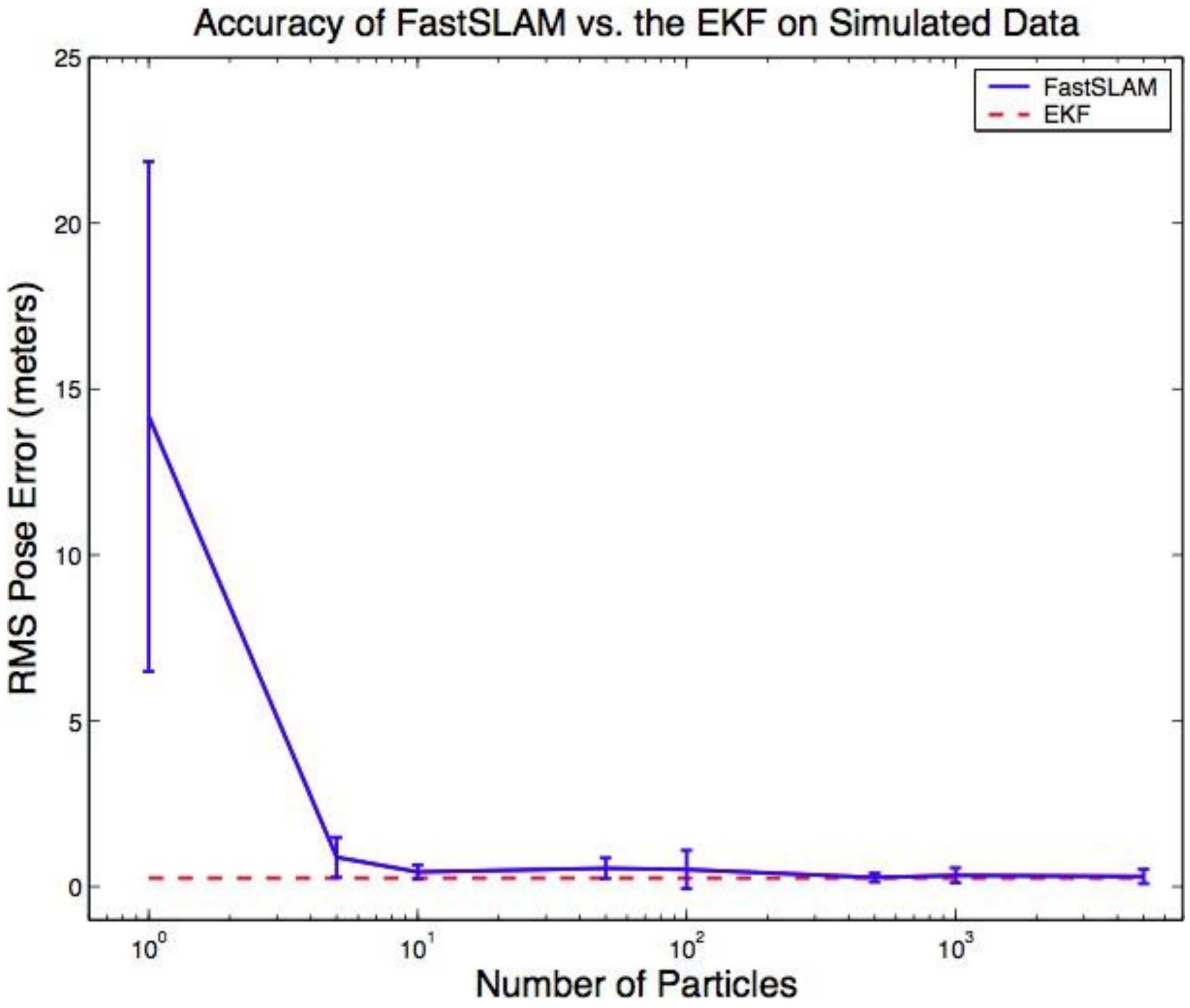
Fast SLAM

Results – Victoria Park

Defining futures

Fast SLAM

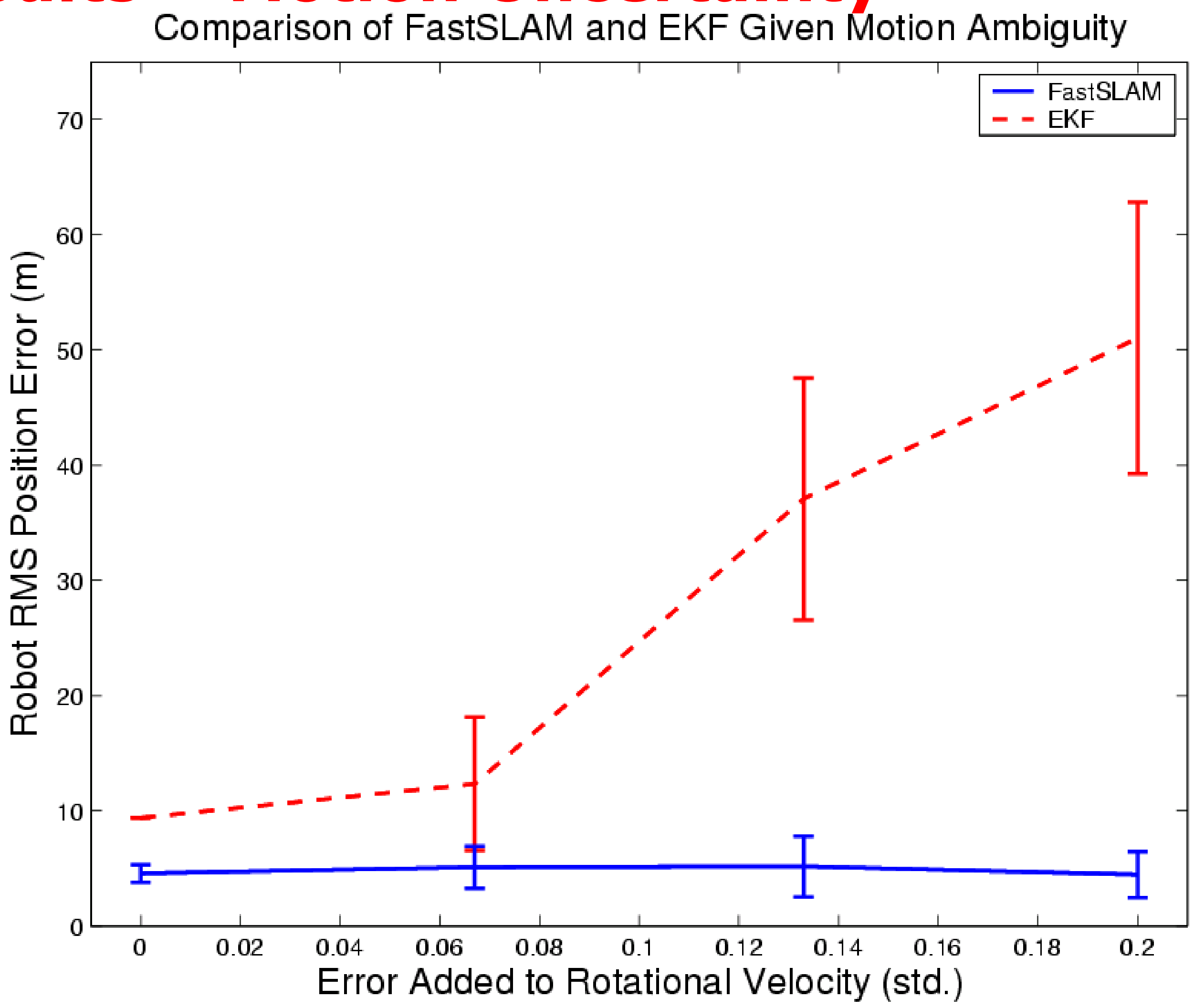
Results – Sample Size



Defining futures

Fast SLAM

Results – Motion Uncertainty



Defining futures

Fast SLAM 1.0 - Summary

- Use a particle filter to model the belief
- Factors the SLAM posterior into low- dimensional estimation problems
- Model only the robot's path by sampling
- Compute the landmarks given the path
- Per-particle data association
- No robot pose uncertainty in the per- particle data association

Defining futures

Fast SLAM

Fast SLAM Complexity – Simple Implementation

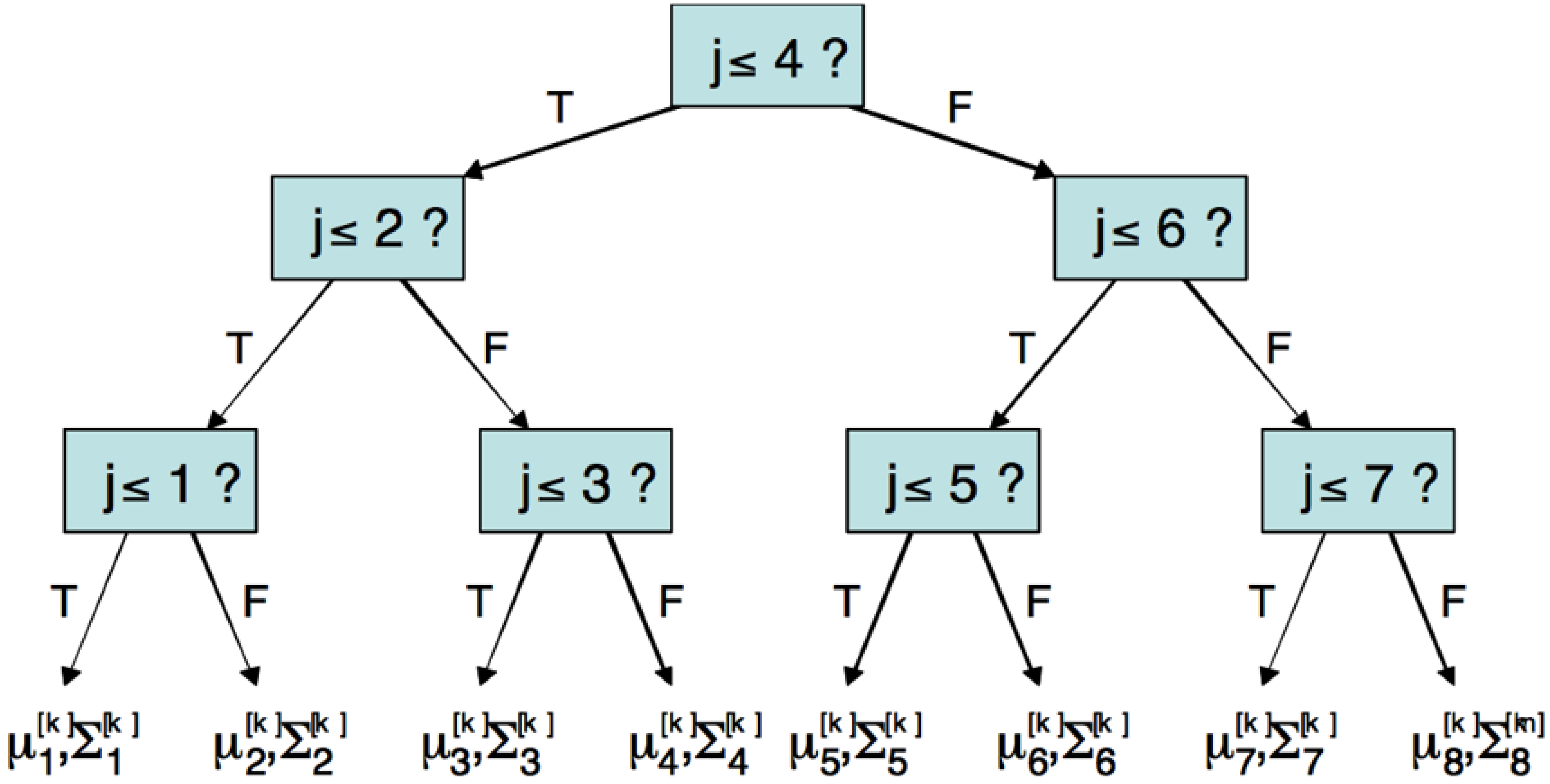
- Update robot particles based on the control $\mathcal{O}(N)$
- Incorporate an observation into the Kalman filters $\mathcal{O}(N)$
- Resample particle set $\mathcal{O}(NM)$

$N = \textit{Number of particles}$
$M = \textit{Number of map features}$

Defining futures

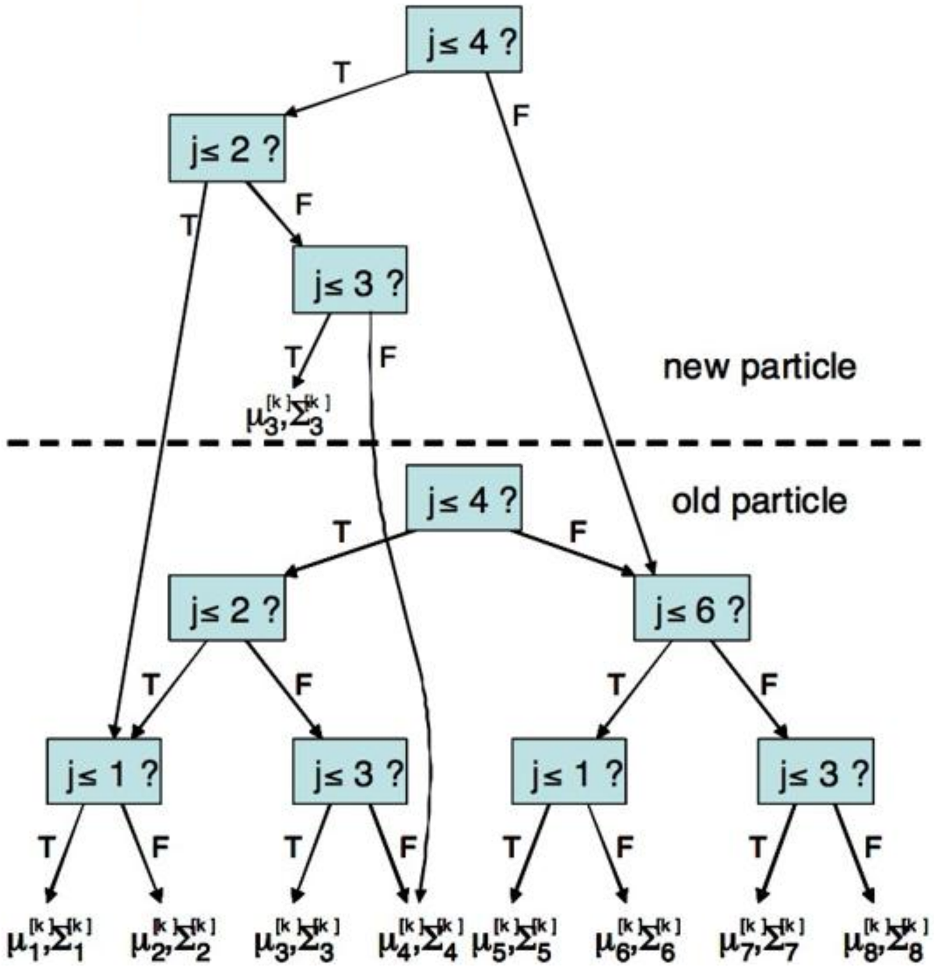
Fast SLAM

A Better Data Structure for Slam



Defining futures

A Better Data Structure for Slam



Fast SLAM

Fast SLAM Complexity

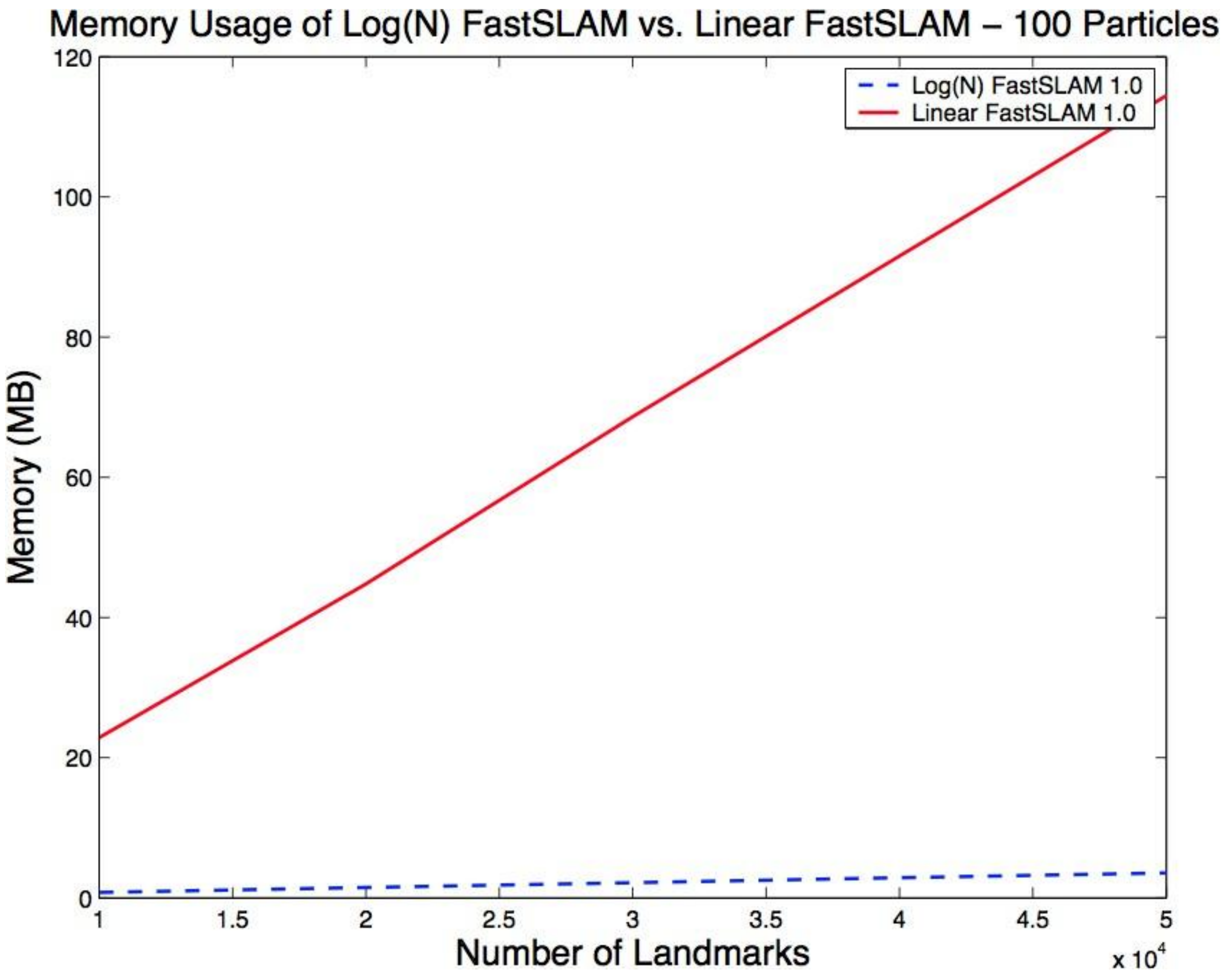
- Update robot particles based on the control $\mathcal{O}(N)$
- Incorporate an observation into the Kalman filters $\mathcal{O}(N \log M)$
- Resample particle set $\mathcal{O}(N \log M)$

$N = Number\ of\ particles$
$M = Number\ of\ map\ features$

Defining futures

Fast SLAM

Fast SLAM Complexity



Defining futures

Fast SLAM

Fast SLAM 2.0

- FastSLAM 1.0 uses the motion model as the proposal distribution

$$x_t^{[k]} \sim p(x_t \mid x_{t-1}^{[k]}, u_t)$$

- Is there a better distribution to sample from?
- FastSLAM 2.0 considers also the measurements during sampling
- Especially useful if an accurate sensor is used (compared to the motion noise)
- FastSLAM 2.0 samples from

$$x_t^{[k]} \sim p(x_t \mid x_{1:t-1}^{[k]}, u_{1:t}, z_{1:t})$$

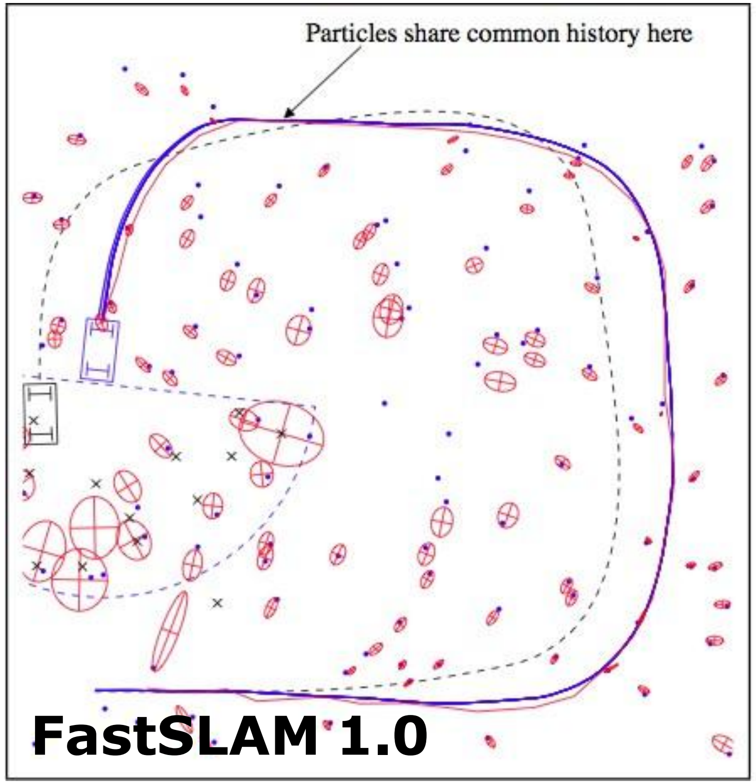
- Results in a more peaked proposal distribution
- Less particles are required
- More robust and accurate
- But more complex…

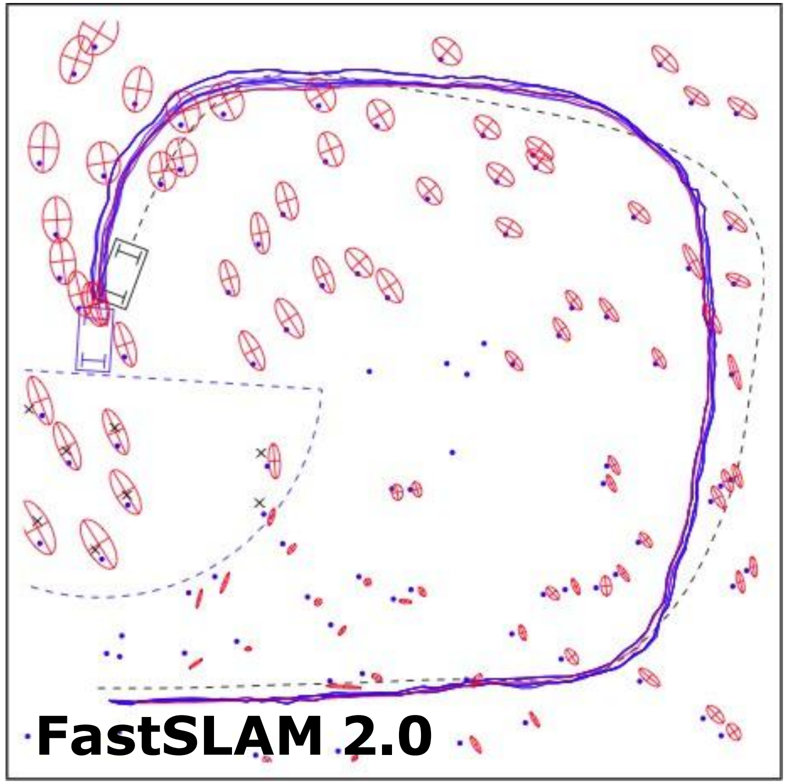
Defining futures

Fast SLAM

Fast SLAM Problems

- How to determine the sample size?
- Particle deprivation, especially when closing (multiple) loops





Fast SLAM

Fast SLAM Summary

- Particle filter-based SLAM
- Rao-Blackwellization: model the robot's path by sampling and compute the landmarks given the poses
- Allow for per-particle data association
- FastSLAM 1.0 and 2.0 differ in the proposal distribution
- Complexity $\mathcal{O}(N \log M)$

Defining futures

Fast SLAM

Fast SLAM Results

- Scales well (1 million+ features)
- Robust to ambiguities in the data association
- Advantages compared to the classical EKF approach (especially with non-linearities)

Defining futures